

Supplemental Table 1: Demographic data of donors proceeding to transplant

	Overall		DBD		DCD	
Characteristic	NMP	SCS	NMP	SCS	NMP	SCS
Age (years)						
Mean $\pm$ SD	53.1 $\pm$ 12.9	52.5 $\pm$ 11.5	56.5 $\pm$ 9.3	54.4 $\pm$ 9.7	35.6 $\pm$ 15.2	38.7 $\pm$ 14.4
Median	54.0	52.0	56.0	53.0	30.5	37.0
Range (Min, Max)	(18.0, 80.0)	(20.0, 79.0)	(40.0, 80.0)	(40.0, 79.0)	(18.0, 66.0)	(20.0, 61.0)
IQR	47.0, 60.0	45.0, 60.0	49.0, 62.0	47.0, 61.0	22.0, 47.0	27.0, 50.0
Male	45.6% (62/136)	52.3% (68/130)	43.0% (49/114)	50.9% (58/114)	59.1% (13/22)	62.5% (10/16)
Race						
White	77.9% (106/136)	75.4% (98/130)	76.3% (87/114)	73.7% (84/114)	86.4% (19/22)	87.5% (14/16)
Black/African American	18.4% (25/136)	20.8% (27/130)	21.1% (24/114)	23.7% (27/114)	4.5% (1/22)	0.0% (0/16)
Other	3.7% (5/136)	3.9% (5/130)	2.6% (3/114)	2.6% (3/114)	9.1% (2/22)	12.5% (2/16)
Diabetes	18.0% (24/133)	11.8% (15/127)	20.7% (23/111)	13.5% (15/111)	4.5% (1/22)	0.0% (0/16)
Smoker	52.7% (69/131)	47.2% (60/127)	53.2% (58/109)	45.9% (51/111)	50.0% (11/22)	56.3% (9/16)
Heavy alcohol history	19.5% (26/133)	22.8% (29/127)	21.4% (24/112)	23.4% (26/111)	9.5% (2/21)	18.8% (3/16)
Illicit drug use history	35.1% (47/134)	41.3% (52/126)	33.0% (37/112)	40.5% (45/111)	45.5% (10/22)	46.7% (7/15)
Body Mass Index (kg/m ²)						
Mean $\pm$ SD	30.2 $\pm$ 7.9	29.4 $\pm$ 6.9	30.5 $\pm$ 8.4	29.8 $\pm$ 6.9	28.9 $\pm$ 5.1	26.6 $\pm$ 6.3
Median	29.2	28.3	29.2	28.5	29.3	25.3
Range	(18.2, 80.7)	(17.3, 51.0)	(18.2, 80.7)	(17.8, 51.0)	(20.7, 37.8)	(17.3, 44.9)
IQR	25.6, 33.8	24.5, 32.6	25.6, 34.6	25.1, 33.1	25.9, 32.6	23.0, 28.0
Diagnosis of Hepatitis C	8.1% (11/136)	3.8% (5/130)	8.8% (10/114)	3.5% (4/114)	4.5% (1/22)	6.3% (1/16)
Cause of Death						

CVA	37.5% (51/136)	44.6% (58/130)	40.4% (46/114)	47.4% (54/114)	22.7% (5/22)	25.0% (4/16)
Hypoxia	2.2% (3/136)	1.5% (2/130)	0.9% (1/114)	1.8% (2/114)	9.1% (2/22)	0.0% (0/16)
Trauma	15.4% (21/136)	15.4% (20/130)	14.0% (16/114)	13.2% (15/114)	22.7% (5/22)	31.3% (5/16)
Anoxia	40.4% (55/136)	34.6% (45/130)	41.2% (47/114)	35.1% (40/114)	36.4% (8/22)	31.3% (5/16)
Other	4.4% (6/136)	3.8% (5/130)	3.5% (4/114)	2.6% (3/114)	9.1% (2/22)	12.5% (2/16)
Donor Risk Index (DRI)						
Mean ± SD	1.6 ± 0.3	1.6 ± 0.3	1.6 ± 0.3	1.6 ± 0.3	1.9 ± 0.4	1.9 ± 0.5
Median	1.6	1.6	1.5	1.6	1.9	1.7
Range	(1.1, 2.8)	(1.0, 3.1)	(1.1, 2.4)	(1.0, 2.3)	(1.4, 2.8)	(1.4, 3.1)
IQR	1.4, 1.8	1.4, 1.8	1.3, 1.8	1.4, 1.8	1.5, 2.1	1.6, 2.0

Supplemental Table 2: Recipient demographics

	Overall		DBD		DCD	
Characteristic	NMP	SCS	NMP	SCS	NMP	SCS
MELD Score - Calculated						
Mean ± SD	19.2 ± 9.5	19.4 ± 8.8	19.6 ± 9.9	19.3 ± 9.3	17.3 ± 7.0	20.3 ± 4.7
Median	18.0	18.0	17.5	18.0	18.0	20.0
Range (Min, Max)	(6.0, 45.0)	(6.0, 42.0)	(6.0, 45.0)	(6.0, 42.0)	(7.0, 35.0)	(12.0, 31.0)
IQR	12.0, 26.0	13.0, 24.0	12.0, 26.0	12.0, 25.0	11.0, 23.0	17.5, 22.0
Age (years)						
Mean ± SD	57.4 ± 10.5	57.2 ± 10.6	57.7 ± 10.4	57.8 ± 10.5	55.8 ± 11.3	52.6 ± 10.3
Median	59.0	60.0	59.5	60.0	58.0	55.0
Range (Min, Max)	(20.0, 76.0)	(21.0, 77.0)	(21.0, 76.0)	(21.0, 77.0)	(20.0, 73.0)	(37.0, 67.0)
IQR	54.0, 64.0	52.0, 65.0	54.0, 64.0	53.0, 65.0	53.0, 62.0	42.0, 59.0
Male	68.4% (93/136)	63.4% (83/131)	69.3% (79/114)	66.1% (76/115)	63.6% (14/22)	43.8% (7/16)
Race						
White	93.4% (127/136)	90.8% (119/131)	93.0% (106/114)	90.4% (104/115)	95.5% (21/22)	93.8% (15/16)
Black or African American	3.7% (5/136)	6.9% (9/131)	3.5% (4/114)	7.0% (8/115)	4.5% (1/22)	6.3% (1/16)
Other	2.9% (4/136)	1.5% (1/131)	3.5% (4/114)	1.7% (1/115)	0.0% (0/22)	0.0% (0/16)
Body Mass Index (kg/m ²)						
Mean ± SD	29.2 ± 5.7	29.5 ± 6.0	29.4 ± 5.8	29.8 ± 5.8	27.8 ± 4.9	27.7 ± 7.5
Smoker	31.9% (43/135)	26.2% (34/130)	32.7% (37/113)	25.4% (29/114)	27.3% (6/22)	31.3% (5/16)
Heavy alcohol history	37.1% (49/132)	33.1% (43/130)	38.7% (43/111)	31.6% (36/114)	28.6% (6/21)	43.8% (7/16)
Etiology of liver disease						
Hepatocellular	81.6%	85.5%	83.3%	87.8%	72.7%	68.8%

	(111/136)	(112/131)	(95/114)	(101/115)	(16/22)	(11/16)
Cholestatic	8.1% (11/136)	6.9% (9/131)	6.1% (7/114)	7.0% (8/115)	18.2% (4/22)	6.3% (1/16)
Vascular	0.7% (1/136)	0.0% (0/131)	0.9% (1/114)	0.0% (0/115)	0.0% (0/22)	0.0% (0/16)
Metabolic	5.1% (7/136)	6.1% (8/131)	5.3% (6/114)	5.2% (6/115)	4.5% (1/22)	12.5% (2/16)
Primary HCC	31.6% (43/136)	29.8% (39/131)	35.1% (40/114)	29.6% (34/115)	13.6% (3/22)	31.3% (5/16)
Other	11.0% (15/136)	13.7% (16/131)	8.8% (10/114)	11.3% (12/115)	22.7% (5/22)	31.3% (4/16)
Re-transplant	0.7% (1/136)	1.5% (2/131)	0.9% (1/114)	1.7% (2/115)	0.0% (0/22)	0.0% (0/16)

Supplemental Table 3: Early Allograft Dysfunction analyzed by donor risk

	Donor Risk Index (DRI)							
	DRI ≤ 1.404948		1.404948 < DRI ≤ 1.622325		1.622325 < DRI ≤ 1.870489		DRI > 1.870489	
	NMP	SCS	NMP	SCS	NMP	SCS	NMP	SCS
EAD	25.0% (10/40)	25.7% (9/35)	17.6% (6/34)	19.4% (7/36)	18.5% (5/27)	15.6% (5/32)	19.2% (5/26)	33.3% (8/24)

Supplemental Table 4: Hospital stay parameters by donor type

	Overall		DBD		DCD	
	NMP	SCS	NMP	SCS	NMP	SCS
ICU stay (days)						
N	134	127	112	112	22	15
Mean $\pm$ SD	4.6 $\pm$ 7.5	4.2 $\pm$ 6.2	4.1 $\pm$ 6.9	4.0 $\pm$ 6.4	7.1 $\pm$ 10.1	5.4 $\pm$ 4.6
Median	2.0	3.0	2.0	3.0	3.5	3.0
Range (Min, Max)	(0.0, 56.0)	(0.0, 59.0)	(0.0, 56.0)	(0.0, 59.0)	(1.0, 47.0)	(1.0, 16.0)
IQR	2.0, 4.0	2.0, 4.0	1.0, 4.0	2.0, 4.0	2.0, 8.0	3.0, 6.0
Total hospital stay (days)						
N	134	127	112	112	22	15
Mean $\pm$ SD	14.3 $\pm$ 17.1	13.3 $\pm$ 18.1	12.2 $\pm$ 12.2	12.5 $\pm$ 18.1	24.9 $\pm$ 30.3	18.8 $\pm$ 18.5
Median	9.0	9.0	9.0	9.0	13.0	10.0
Range (Min, Max)	(3.0, 115.0)	(4.0, 185.0)	(4.0, 88.0)	(4.0, 185.0)	(3.0, 115.0)	(6.0, 57.0)
IQR	7.0, 14.0	7.0, 13.0	6.0, 13.0	7.0, 12.5	8.0, 21.0	6.0, 28.0
Total ventilation days post-implant (days)						
N	132	124	110	109	22	15
Mean $\pm$ SD	1.7 $\pm$ 5.3	0.9 $\pm$ 1.3	1.6 $\pm$ 5.7	0.8 $\pm$ 1.4	1.9 $\pm$ 3.0	1.2 $\pm$ 0.8
Median	1.0	1.0	1.0	0.0	1.0	1.0
Range (Min, Max)	(0.0, 56.0)	(0.0, 9.0)	(0.0, 56.0)	(0.0, 9.0)	(0.0, 12.0)	(0.0, 2.0)
IQR	0.0, 1.0	0.0, 1.0	0.0, 1.0	0.0, 1.0	0.0, 2.0	1.0, 2.0
Postop renal replacement therapy	15.7% (21/134)	15.7% (20/127)	16.1% (18/112)	15.2% (17/112)	13.6% (3/22)	20.0% (3/15)

Supplemental Table 5: Device malfunctions

	Type of malfunction	Timing of malfunction	Description of malfunction	Action taken	Impact	Recipient adverse event
1	Disposable set	During set up	Broken syringe in set	Replaced	None	None
2	Disposable set	During set up	Crack in liver bowl air filter	Replaced	None	None
3	Disposable set	During cold flush after NMP	Leak in cold flush set	None	None	None
4	Disposable set	During set up	Leak from Luer lock connector	Disposable set replaced	None	None
5	Disposable set	During set up	Leak from Luer lock connector	Leak repaired	None	None (patient died from recurrent HCC)
6	Disposable set	During set up	Jammed Luer lock connector	Disposable set replaced	None	None (patient died from secondary hemorrhage)
7	Disposable set	During NMP, before transport	Leak from Luer lock connector	Switched to SCS	NMP not performed	Liver not transplanted for recipient reason
8	Device calibrator	During set up	Blood gas calibrator fault	Switched to SCS	NMP not performed	EAD and biliary stricture
9	Disposable set	During set up	No bile salts in perfusion pack	Bile salts replaced	None	None
10	Device	During set up	Damaged arterial pressure sensor	Switched to SCS	NMP not performed	EAD
11	Device	During set up	Malfunction of blood gas analyzer	Device recalled	DCD donor – did not proceed	None
12	Device	During NMP, prior to transport	Transient drop in O ₂ delivery	Software restart	None	None
13	Disposable set	During set up	Leak from Luer lock on IVC line	Disposable set replaced	None	None
14	Device	During NMP, prior to transport	Unable to fully charge battery	Mains power used (as per IFU)	None	None

Supplemental Table 6: causes of graft loss

	Normothermic perfusion		Static cold storage	
	Graft loss cause	Retransplant	Graft loss cause	Retransplant
1	Cholangiopathy	10 months	Primary non-function	Day 3
2	Hepatic artery thrombosis	10 weeks	Hepatic artery thrombosis	Day 11
3	Primary graft dysfunction	Day 9	Cholangiopathy	4 months
4	Hepatic artery thrombosis	Day 35		

Supplemental Table 7: Analysis of patient deaths

	0 Day	30 Days	3 Months	6 Months	12 Months
NMP					
At risk ¹	136	133	131	127	120
Subjects with events	0	3	3	6	10
Event free ²	100.0%	97.8%	97.8%	95.5%	92.5%
Standard error	0.0%	1.3%	1.3%	1.8%	2.3%
95% CI	(100.0%, 100.0%)	(93.3%, 99.3%)	(93.3%, 99.3%)	(90.3%, 98.0%)	(86.6%, 95.9%)
SCS					
At risk ¹	130	127	126	125	121
Subjects with events	0	1	2	2	4
Event free ²	100.0%	99.2%	98.4%	98.4%	96.6%
Standard error	0.0%	0.8%	1.1%	1.1%	1.7%
95% CI	(100.0%, 100.0%)	(94.7%, 99.9%)	(93.9%, 99.6%)	(93.9%, 99.6%)	(91.3%, 98.7%)
¹ Number of subjects at risk at day 0 or the beginning of visit windows (23, 76, 166, 335 days from procedure).					
² Estimates made at day 0 or the scheduled visit days (30, 90, 180, 365 days from procedure).					

Supplemental Table 8: causes of death

	Normothermic perfusion		Static cold storage	
	Cause of death	Postop time	Cause of death	Postop time
1	Pneumonia	5 months	Post-reperfusion cardiac failure	Day 1
2	Secondary hemorrhage	Day 9	Recurrent HCC	11 months
3	Recurrent HCC	9 months	Sepsis, lymphoma	2 months
4	Intracranial hemorrhage	Day 1	Posterior Reversible Encephalopathy Syndrome (PRES)	8 months
5	Status epilepticus	Day 15		
6	Cardiomyopathy	5 months		
7	Pneumonia + CVA	4 months		
8	Recurrent HCC	9 months		
9	Hepatic arterial thrombosis (graft)	10 months		
10	Pneumonia	6 months		